

LLMOps & Evaluation Cheat Sheet

Baseline written 2026-10-01: vLLM 0.30.0, SGLang 0.5.21, KServe 0.21.0, LiteLLM 1.103.2, MLflow 3.16.1, Langfuse 4.49.0 (SDK 4.16.0), LangSmith SDK 0.14.2 + openevals 0.2.0 + agentevals 0.0.9, Arize Phoenix 20.18.0, RAGAS 0.4.3, DeepEval 4.2.7, OpenTelemetry Python 1.45.0, OpenLLMetry 0.62.4, OTel GenAI conventions at Development stability (gen-ai-dev/1.42.0-dev on semconv 1.44.0), Spring AI 2.0.1 (Spring Boot 4.1.x, Java 21), LangChain 1.4.3 (Python 3.12), Prometheus 3.15.0, Grafana 13.2.3. Snippets are illustrative, not compiled; no benchmark numbers quoted.

LIFECYCLE AND LATENCY

Loop: **Build** (prompts, retrieval, tools) → **Evaluate** (golden set, judges) → **Deploy** (gateway, shadow, canary) → **Observe** (traces, metrics, cost) → **Improve** (feedback, drift, new cases) → Build; sampled live traffic flows back into Evaluate.

- Level 0 ad hoc:** prompts in code or notebooks, judged by eye, no traces, cost from the invoice.
- Level 1 managed:** prompts and models versioned and pinned, golden set gates CI, traces and golden-signal metrics, gateway with keys and budgets, runbooks.
- Level 2 optimised:** calibrated sampled online judging, feedback into the golden set, shadow/canary/A/B, drift monitoring, per-tenant cost, metric owners, audit trail.

METRIC CATALOGUE

Metric	Use / formula
pass@k	code, structured output, tool calls: $E[1 - C(n-c,k)/C(n,k)]$; $k=1$ gives c/n ; report n
Exact match, F1	short gold answers, extraction
BLEU	$BP \cdot \exp(\sum w_i \log p_i)$; $BP=1$ if $c>r$ else $e^{1-r/c}$; translation, corpus level
ROUGE	n -gram / LCS recall; summaries, long form
BERTScore	contextual embeddings vs reference; paraphrase tolerant
Embedding sim.	$\cos(u,v) = u \cdot v / (u v)$; cheap screen, drift
Perplexity	$\exp(-1/T \sum \log p(x_i x_{<i}))$; open weights, quantisation regressions
Rubric judge	open-ended quality, groundedness, tone
RAG triad	RAGAS/DeepEval: faithfulness, context precision, answer relevance (reference-free)

- Component evals vs system evals; attribute a failure to a stage (retrieval, generation, tool).
- Agents: final response, single step, trajectory; tool-call accuracy, task completion, repeat-run reliability, cost per task.

JUDGE RUBRIC (DOCSASSISTANT)

Criterion	Inputs	Scale
Groundedness	question, context, answer	binary; any unsupported claim fails
Correctness	question, answer, reference	1-4 anchored (4 same facts, 1 contradicts)
Refusal quality	question, context, answer	binary; refuse only if context lacks it
Tone/format	answer	binary; style guide

One criterion per judge; binary or small anchored scales.

JUDGE BIAS CHECKLIST

- Position:** run both orders, count only agreeing pairs; report flip rate = $(1/n) \sum 1[J(a,b) \neq J(b,a)]$ mapped back.
- Verbosity:** rubric says length is no virtue; compare winner/loser lengths.
- Self-preference:** judge from a different model family.
- Leniency/compression:** binary strict criteria, known-bad examples.
- Style/authority:** judge against context and reference.
- Non-determinism:** temperature 0, seed, cache by hash(prompt version, inputs).
- Calibrate vs two human labellers: kappa (weighted if ordered), confusion matrix; correct live aggregates for judge error; pin judge versions and re-validate.

COMPARATIVE EVALUATION

- Elo: $E_A = 1/(1+10^{(R_B-R_A)/400})$; 400 points = 10:1 odds.
- Bradley-Terry: $r_i = r_0 + 400 \log_{10} \pi_i$, fitted on all battles at once.
- Wilson interval for win rates; bootstrap for ratings; blind assignment in an internal arena.

Eval-Set Rules

- $n \geq z^2 p(1-p)/e^2 \approx 0.96/e^2$: ~100 items = ± 10 pts, ~400 = ± 5 . Rule of three: 0 failures in n trials gives upper bound ~3/n.
- Written taxonomy; minimum items per stratum; unanswerable, out-of-scope, multi-turn, adversarial items present.
- Real traffic scrubbed of personal data, provenance kept; synthetic items from a different model family, human reviewed, deduplicated.
- Two labellers on a shared slice, record kappa, drop ambiguous items.
- Fast gate set (smoke subset by stable tag) plus larger selection/nightly set; held-out split untouched by tuning.
- Git as source of truth, manifest with version and hash; failures from production flow back in.

CI GATE

Layers, cheapest first: deterministic checks → retrieval metrics → judge metrics (cached) → nightly: large set, pairwise vs release, trajectories.

Wilson lower bound $p_{l0} = (p + z^2/2n - z/(p(1-p)/n + z^2/4n^2)) / (1 + z^2/n)$, $z = 1.96$. Paired exact sign test: $p = \sum_{i=b..b+c} C(b+c,i)/2^{b+c}$.

```
FL00R=0.80 ALPHA=0.05 MIN_DROP=0.03
if lower < FL00R: fail("flood")
if p < ALPHA and drop > MIN_DROP:
    fail("regression vs baseline")
sys.exit(1) on any failure
```

- False alarm with r repeat runs: $1-(1-q)^r$ ($q=.05$, $r=3$: ~14%). Majority of runs; re-run only failures, never retry-until-green.
- Pin temperature 0, seed, dated model id; report unstable items; update baseline deliberately.

ONLINE EVALUATION

- Never judge on the request path: queue/worker reads the stored trace, writes the score back.
- Sample a few percent, deterministic by hash of trace id; oversample no-citation, low retrieval score, thumbs-down, long chats, new tenants.
- $n \geq z^2 p(1-p)/e^2$: $e=.03$ at 95% needs ~1,070 judged answers per window per slice; small tenants need a higher rate.
- Explicit (thumbs) and implicit (regenerate, follow-up) feedback; watch bias and degenerate loops; labelled failures feed the golden set (flywheel); Thompson sampling for bandits.

SERVING METRICS AND SCALING

- $TTFT = t_q + t_0$ (queue, gateway, prefill, first decode).
- $TPOT = (t_q - t_1)/(N-1)$; $e2e = TTFT + (N-1) TPOT$; $ITL =$ per-gap sample.
- Goodput = $(1/W) \sum 1[TTFT_i \leq \tau_{TTFT} \wedge TPOT_i \leq \tau_{TPOT}]$; size capacity by it.
- $MBU = (M_{weights} + M_{KV,read}) r_{tok} / B_{peak}$.
- Engine /metrics: requests waiting (queue), running (batch), KV-cache use, $TTFT/TPOT/e2e$ histograms, prefix-cache hits, token counters; vllm: names change by release, check them.
- Autoscale (KEDA) on queue depth (~4/replica) and KV-cache use (~0.8), not CPU; fast up, slow down, warm floor, cooldown 600 s.
- Prefix caching: stable prompt prefix first. TGI 3.3.7 in maintenance mode: prefer vLLM or SGLang; KServe on Kubernetes.

RELEASE PATTERNS

Pattern	Who sees candidate
Shadow	nobody; mirrored copy, own budget and sampling; no user signals
Canary	small sticky share by user/conversation hash; abort rules set first
A/B	two stable groups; long enough for statistics
Blue-green	all users at once; fast reversal

- Gateway (LiteLLM): routing, fallbacks, virtual keys with budgets; master key stays with platform. Version prompts and models apart from code; pin dated model ids, never aliases.
- Rollback ladder: flag off → prompt label back → gateway weight → previous image → index snapshot. Keep previous version deployable; rehearse.
- Tag every request with its variant; canary for operational health, shadow for quality.

OTEL GEN_AI.* ATTRIBUTES

- `gen_ai.operation.name`: chat, embeddings, retrieval, `invoke_agent`, `execute_tool`; span name = operation + model.
- `gen_ai.provider.name` (was `gen_ai.system`), `gen_ai.request.model`, `gen_ai.response.model` (differs under aliases = drift evidence).
- `gen_ai.usage.input_tokens`, `gen_ai.usage.output_tokens`, `gen_ai.response.finish_reasons`.
- `gen_ai.request.temperature`, `gen_ai.request.max_tokens`, `gen_ai.agent.name`, `gen_ai.conversation.id` (pseudonymous), `gen_ai.tool.name`, `gen_ai.tool.call.id`, `error.type`.
- Metrics: `gen_ai.client.token.usage` (`gen_ai.token.type` input/output), `gen_ai.client.operation.duration`, `gen_ai.server.time_to_first_token`, `gen_ai.server.time_per_output_token`.
- Content capture is opt-in. Span rate = $R \cdot s \cdot p$; metrics unaffected by trace sampling. MCP: `mcp.method.name`, `mcp.session.id`. Everything is Development status: pin the snapshot.

PLATFORM COMPARISON

Platform	Licence	Self-host
LangSmith	proprietary; SDKs MIT	enterprise option
Langfuse	MIT except ee/	Compose, Helm; Postgres, ClickHouse, Redis, blob
MLflow 3	Apache-2.0	server + DB + artifacts
Phoenix	Elastic 2.0 (source-available)	one container; SQLite/Postgres
Grafana/Tempo/Loki	AGPL-3.0 core	yes; no built-in evals or prompts

All but Grafana stack bring evals, datasets and prompt management with their own data models; the portable layer is the OTLP trace.

ALERT RULES

Golden signals + token usage, cost, quality, cache hit rate. Labels: model, route, tier, outcome (never user or trace id). Percentiles from histograms.

```
ErrorBudgetFastBurn for 2m page
ErrorBudgetSlowBurn for 15m ticket
TTFTHigh p95_5m > 2 for 10m page
JudgeScoreDegraded faithfulness < 0.85 for 2h tick
et
VllmQueueBuilding avg waiting[5m] > 10 for 10m
CacheHitRatioDropped for 30m ticket
LlmSpendRateHigh rate>3600 > ceiling 10m page
TokensPerRequest > 3x 1d avg for 15m
```

- Drift: $PSI = \sum (q_i - p_i) \ln(q_i/p_i)$; JSD bounded by $\ln 2$; Wilson-bound and control-chart rules on judge scores; scheduled golden-set runs vs live endpoint.
- Four silent changes: vendor model update, input drift, retrieval/data drift, output drift.
- page → on-call; ticket → queue; each alert has a runbook; hard gateway limits back them.

COST FORMULA AND LEVERS

$C_{req} = n_{in} p_{in} + n_c p_c + n_{out} p_{out}$
 $C_{month} = R(1-h)((1-b) + b \cdot d) \text{mean}(C_{req}) + C_{fixed}$

- Prompt caching** (n_c): stable prefix first; saved share $n_c/(n_{in}+n_c)$.
- Batch APIs** (b, d): CI, regression runs, offline judging.
- Routing/cascade** (prices): $E[C] = C_{small} + q \cdot C_{large}$; pays off if $q \cdot C_{large} < (1-q) C_{large} - C_{small}$.
- Context trimming** (n_{in}): fewer passages, windowed history, strip boilerplate, cap output.
- Semantic cache** (h): tuned threshold, TTL, partition key, measure false hits.
- Gateway budgets (max_budget, tpm, rpm) per key/team; per-tenant attribution.
- Self-host: $p_{self} = F/(T \cdot u \cdot t_{month})$; break-even $V^* = F/p_{hosted}$.
- Every cost cut goes through the eval gate.

GOVERNANCE CHECKLIST

- Named owner per metric (product, application, platform/SRE, data owner); owner receives its alert.
- Runbooks: hallucination incident, provider outage; severity and communication plan.
- Audit trail: who approved which prompt, model or label, justified by which eval run.
- Prompt registry, versions and approvals; model catalogue and allow-list; model and system cards; reproducible release tuple.
- Data ops: preprocessing pipeline, lineage, personal-data handling, index freshness SLAs, backups with restore tests.
- Self-assess levels 0-2; schedule game days; revisit when system or obligations change.